



GREY BANNER ALERT



mainroads
WESTERN AUSTRALIA

SHW Serious Incident (SI) – Final Notice

Date of Incident:	19/7/2022	Time of Incident:	1330hrs
Region / Directorate:	PTS	Main Roads or Contractor:	Contractor
EQSafe Event Type:	Injury / Illness	EQSafe Incident Number:	27663
Actual Consequence:	Moderate	Potential Consequence:	Major

Incident Short Description:	Burst high pressure hydraulic hose resulting in Lost Time Injury (LTI)
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Executive Summary:

On 19th July 2022, a soil mechanics team was conducting a drilling operation for the Mandurah Estuary Bridge project. Soil mechanics had two drill rigs in the area, one drilling on land and the other working over the water on a barge. The incident occurred with the rig DR002, which was working over the water.

Worker A, B and C started the shift at 6:30am. Worker A was the supervisor for the team. Worker A conducted the drill prestart at 7:00am and started the drilling operation around 7:20am. Worker B and C were on site and engaged in other supporting activities for collecting good quality samples for the client. The drilling operation was continued by Worker A until 12:00pm, and at 12:00pm Worker A handed over the drilling operation to Worker B.

From 12:00pm Worker B started drilling and was assisted by Worker C. The drilling continued as normal until 1:30pm. Around 1:30pm Worker B had finished his drill run down to 31 metres and was flushing the hole to lift all the drill cuttings out of the hole. He was using 100% water flow to complete this action. The water hose to the drill head burst at this time, and the water and drill mud splashed through the hole. Worker B moved his safety glasses from the position on his face and looked at the hose. Suddenly, water from the hole struck into Worker B's left eye and face.

Worker C heard a noise and approached Worker B. The client representative on site pressed the emergency button of the rig to shut it down. They moved to the land from the barge in a boat and Worker C took Worker B to the hospital in company vehicle. On the way to the hospital, Worker B called his supervisor Worker A as well as the Managing Director around 1:50pm. The injured person (Worker B) reached the hospital at 2:12pm.

Incident Photos:



Burst Hose



Eye Injury

Contributing Factors:

- Wear and tear of the water hose.
- First aid not given on site / eyes of the injured person was not flushed on site.
- The Injured Person moved their safety glasses from their face.

Corrective Actions:

- Comprehensive inspection shall be conducted on the company drill rigs to identify the condition of hydraulic and pneumatic hoses.
- Conduct a periodic inspection to check all pressure release valves are correctly set.
- Implement periodic water delivery systems checks, including burst sock conditions, pressure release valve pressure, static testing of hoses and whip check condition.
- Provide training to all staff on selecting, using, and maintaining PPE in a proper manner.
- Ensure at least two trained first aid providers on site.

Additional Issues/Learnings:

- Comprehensive periodic inspections are very important to avoid unplanned maintenance to the equipment. The water and pneumatic hoses may have wear and tear over the course of time, this needs to be considered while conducting comprehensive inspections.
- Drill rigs are working on different stored energy. Even if different safety controls are installed in the rig machine, there is a potential to have sudden release. So, no worker is allowed to ease the proper use of PPE.
- First aid is crucial treatment for an injured person in the event of an accident or emergency situation until help arrives. All sites need to have at least more than one trained first aid personnel.

Distribution of Notice:

Main Roads Personnel and Contracting Organisations